Taxora: An AI-Based Accounting and Compliance Platform for SMEs and Chartered Accountants

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Abstract -

Small and medium-sized businesses (SMEs) in India continue to rely on manual financial management practices involving extensive paperwork and poorly organized records. These limitations often lead to delays in tax compliance, increased operational costs, and a higher risk of documentation errors.Chartered Accountants (CAs) handle large volumes of client invoices and financial data that require continuous manual verification. Compliance-related tasks consume a significant portion of their time, reducing productivity and increasing the possibility of inaccuracies in reporting and filing.This study introduces Taxora, an AI-driven accounting and compliance platform that digitizes financial workflows and enables secure, role-based collaboration between SMEs and Chartered Accountants. The system automates document handling and provides decision support through integrated tools. Taxora employs a fine-tuned LayoutLMv3 model to extract structured information from invoices by analyzing both textual and spatial features. The extracted data is processed through a compliance engine based on Indian tax regulations to generate audit-ready GST summaries and reports. The platform is implemented using Django and PostgreSQL with an emphasis on automation, privacy, and scalability.The system significantly reduces manual data entry and improves extraction accuracy, resulting in faster processing, improved operational efficiency, and more reliable compliance for SMEs and Chartered Accountants.

Keywords -

Invoice Automation , SME Accounting, GST Compliance, OCR and NLP, Financial Analytics, AI in Accounting

I. Introduction

The growth of SMEs in India highlights the urgent need for smarter and more reliable accounting systems. Small and Medium Enterprises (SMEs) play a vital role in the Indian economy, contributing significantly to employment generation and overall economic growth. Despite their importance, a large proportion of SMEs continue to rely on manual or semi-digital accounting practices. This dependence often leads to inefficiencies in invoice handling, fragmented expense tracking, and increased risk of non-compliance with tax regulations [3]. Managing invoices across heterogeneous formats, interpreting evolving GST requirements, and maintaining timely coordination with Chartered Accountants (CAs) remain persistent challenges for business owners, particularly those with limited digital infrastructure [9].

In recent years, the increasing frequency of regulatory updates, including GST amendments, e-invoicing mandates, and digital reporting requirements, has further intensified the compliance burden on SMEs. Manual accounting systems struggle to adapt quickly to these changes, increasing the likelihood of delayed filings and statutory penalties. This evolving regulatory environment necessitates intelligent systems that can dynamically support compliance while minimizing human intervention [11].

From the professional perspective, CAs are responsible for managing multiple SME clients concurrently, each submitting financial documents in varied and unstructured formats. Manual verification, repetitive data entry, and continuous follow-ups for missing documents consume a substantial portion of their working hours. These operational burdens reduce the time available for higher-value advisory services, financial analysis, and compliance planning [2]. Existing accounting and tax software solutions offer partial digitization but often lack intelligent automation and seamless collaboration mechanisms between SMEs and CAs [11].

The problem formulation for this study was informed by informal discussions with SME owners and practicing Chartered Accountants, which helped identify recurring challenges related to invoice management, manual data entry, and compliance delays in real-world accounting workflows. These practitioner insights provided practical context to the problem and highlighted gaps between existing accounting tools and actual operational needs.Therefore, there is a strong need to develop an intelligent digital system that simplifies accounting processes, automates invoice management, and ensures accurate tax compliance for SMEs [1].

II. Literature Survey

This section presents a comprehensive review of existing research related to artificial intelligence, machine learning, robotic process automation, and digital finance solutions applied to invoice processing and accounting systems. The review examines document understanding techniques, automation frameworks, and SME-oriented financial platforms to identify technological advancements and research gaps relevant to the proposed system.

AI and ML have been increasingly applied to automate invoice processing, addressing challenges in handling unstructured financial documents and improving efficiency in accounting workflows. One study developed an intelligent invoice processing system using Tesseract OCR to extract key data from invoices, demonstrating enhanced accuracy in digitizing financial records for compliance purposes [1]. Similarly, a framework combining RPA with multimodal approaches was proposed to understand and process unstructured financial documents, enabling automated extraction of critical fields like amounts and vendor details [4]. Another research utilized computer vision techniques for intelligent automation of invoice parsing, focusing on robust handling of varied document formats to reduce manual intervention [5].

Advanced deep learning models have shown promise in extracting information from diverse invoice layouts, overcoming limitations of traditional rule-based systems. A method employing YOLOv11 and PaddleOCR was introduced for automated invoice field detection and data extraction, achieving high precision in identifying elements such as tax components and totals [8]. In parallel, information extraction techniques tailored for different invoice layouts were explored, using imaging science to adapt to structural variations and improve data accuracy [10]. Pre-training models like LayoutLMv3 have further advanced document AI by unifying text and image masking, enabling layout-aware understanding that captures spatial relationships in financial documents for better field extraction [6].

Robotic Process Automation (RPA) and AI innovations are transforming the role of accountants, shifting focus from repetitive tasks to strategic oversight in finance operations. Research highlighted how process robotics technologies are reshaping accounting practices, automating routine workflows to enhance productivity and decision-making [2]. Prospects for AI in accounting were examined, emphasizing its potential to streamline tasks like data verification and reporting while maintaining human expertise for complex judgments [11].

Digital finance tools and platforms are critical for SMEs facing challenges in financial management, compliance, and operational efficiency. A study analyzed factors influencing SME listing on

platforms, identifying barriers like manual processes and the need for automated solutions to improve accessibility [3]. Enhancing SME performance through digital finance was investigated, underscoring the importance of accessibility and literacy in tools that support compliance and growth [9]. Additionally, reusable Django libraries were discussed for developing standalone apps, providing a foundation for secure, scalable backend systems in financial platforms [7].

While AI, ML, and RPA have significantly advanced invoice processing and automation in accounting, particularly through layout-aware models and computer vision, addressing SME-specific challenges like unstructured documents and CA collaboration requires integrated, privacy-focused platforms. This highlights the need for further research on ethical implementations, bias mitigation in diverse Indian tax contexts, and clinically viable solutions like Taxora to bridge manual gaps in compliance workflows.

III. Methodology

The system begins with the upload of invoices by SMEs in either PDF or image format through the platform interface. Once the document is received, Optical Character Recognition (OCR) is applied to convert the content into machine-readable text [1]. After text extraction, a layout-aware model based on a fine-tuned LayoutLMv3 architecture analyzes the document structure to locate and extract important financial details such as the invoice number, supplier information, GST values, and total payable amount [6].

Following extraction, the identified data is checked for correctness and completeness. If any required fields are missing or inconsistencies are found, the record is sent back for correction and verified again. After successful validation, the system calculates the applicable GST and other tax amounts using the extracted financial information [5].

The verified data is then securely saved in the database, along with activity logs for audit purposes. [7] Using this stored information, the platform prepares compliance-ready outputs, including GST summaries and financial statements. These outputs can be accessed by authorized users, shared with clients, and submitted through the integrated GST filing module [10].

At every stage, the platform maintains controlled access based on user roles and allows Chartered Accountants to review and approve records when necessary. Automated alerts and secure communication features support smooth coordination between SMEs and CAs, helping maintain accuracy, compliance, and overall efficiency in the accounting process [2].

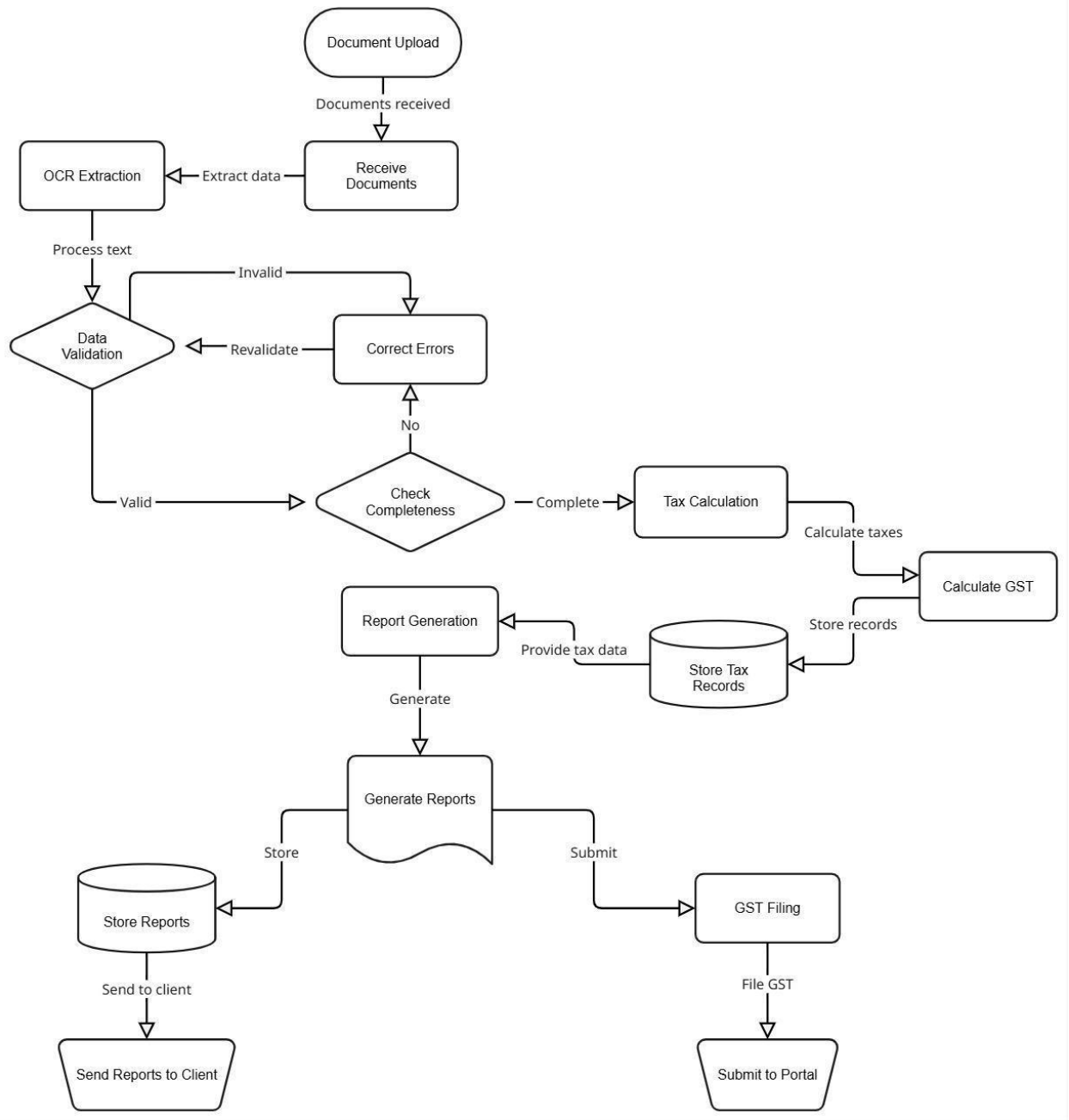

Fig 1. Data Flow Diagram of Taxora

IV. Result

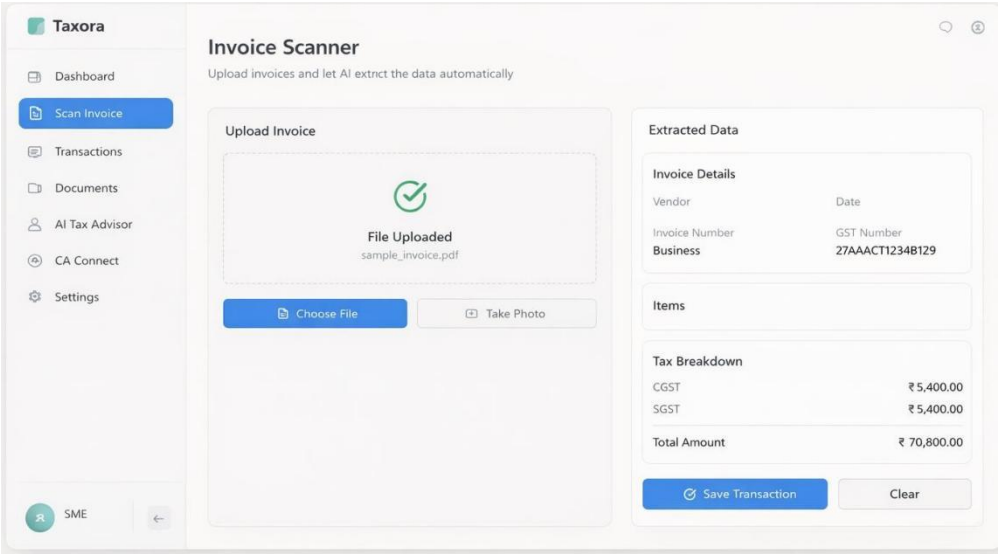

Fig 2. Invoice scanner

This figure shows the invoice scanning interface where uploaded documents are processed using a layout-aware extraction model. Key financial fields are automatically detected and displayed alongside the original invoice for verification.

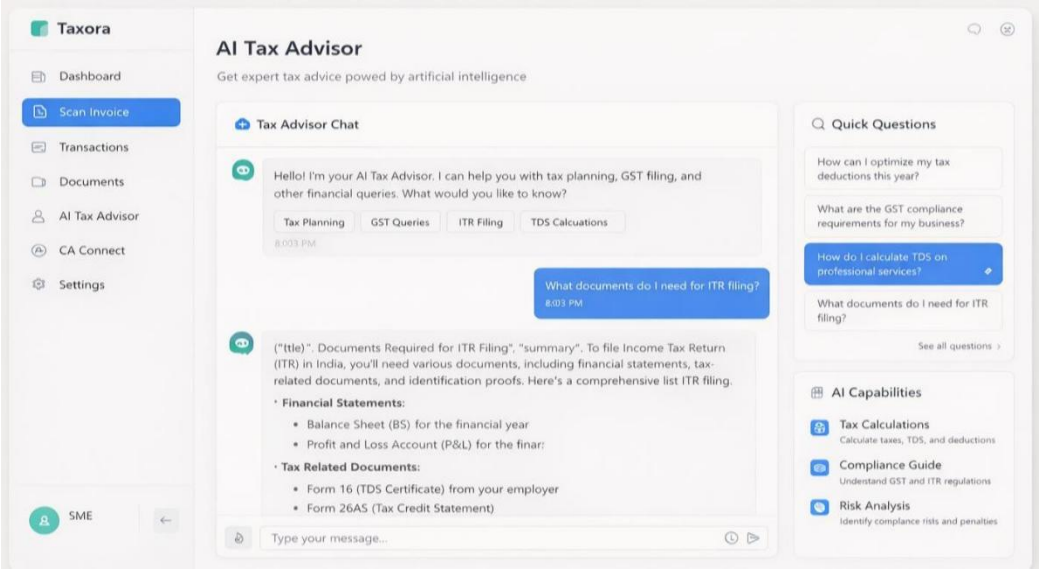

Fig 3.AI Tax Advisor

The AI tax advisor interface demonstrates how users can query finance and tax-related information in natural language. The system provides contextual responses and compliance guidance based on extracted financial data.

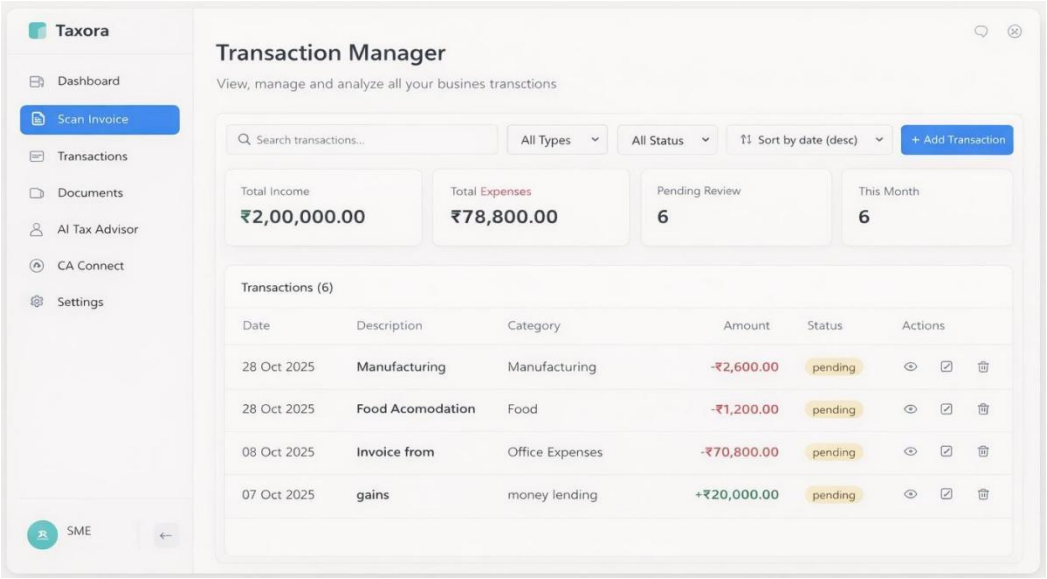

Fig 4. SME Transaction Manager

This figure illustrates the transaction management module, displaying a chronological list of income and expense records. The module enables SMEs to review financial history and track business transactions efficiently

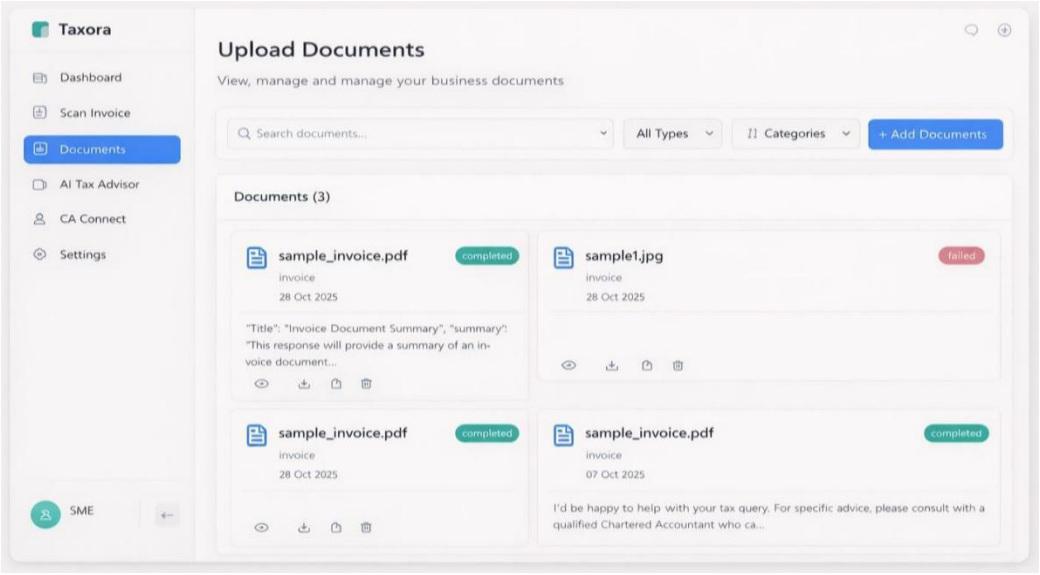

Fig 5. SME Document Management

The document management interface allows SMEs to upload, store, and organize financial documents securely. All records are indexed for quick access and retrieval during audits or compliance checks.

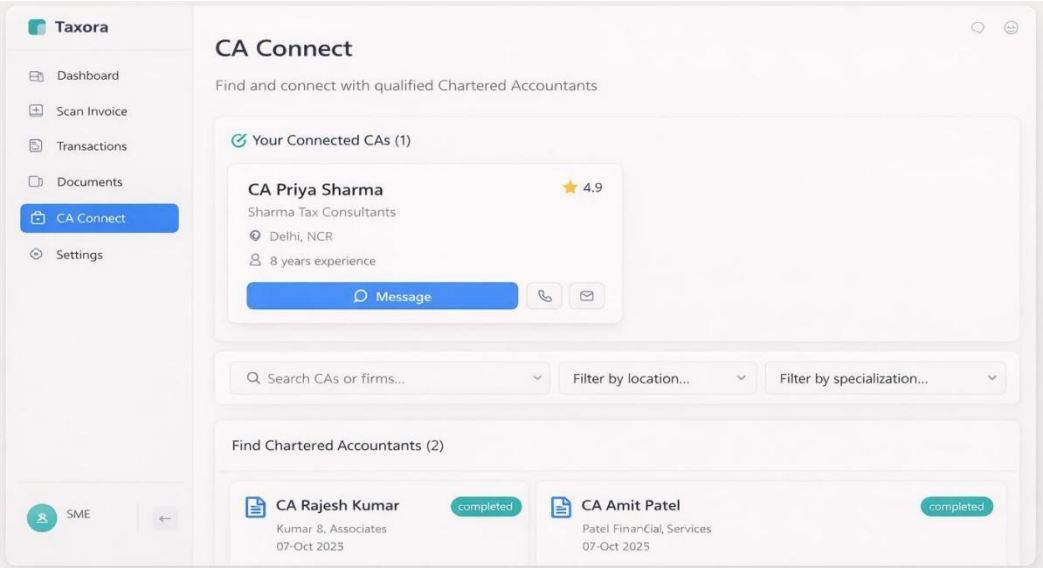

Fig 6.CA Connect Module(SME View)

This output highlights the CA connection module where SMEs can discover, invite, and collaborate with Chartered Accountants. It facilitates structured communication and secure sharing of financial records.

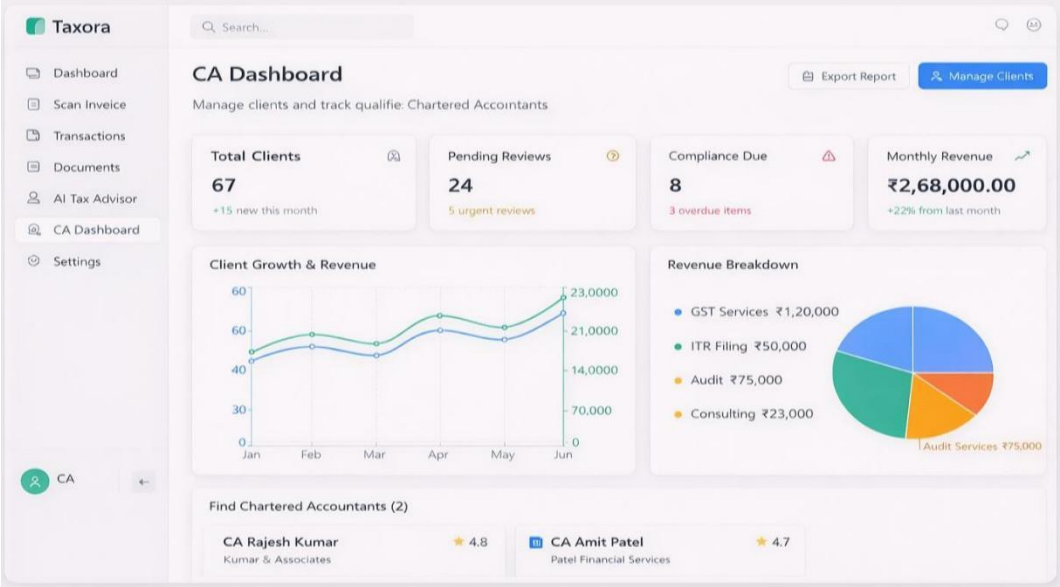

Fig 7 .CA Dashboard

The CA dashboard provides a unified overview of multiple clients, including compliance status, pending tasks, and financial summaries. It supports efficient monitoring and professional decision-making.

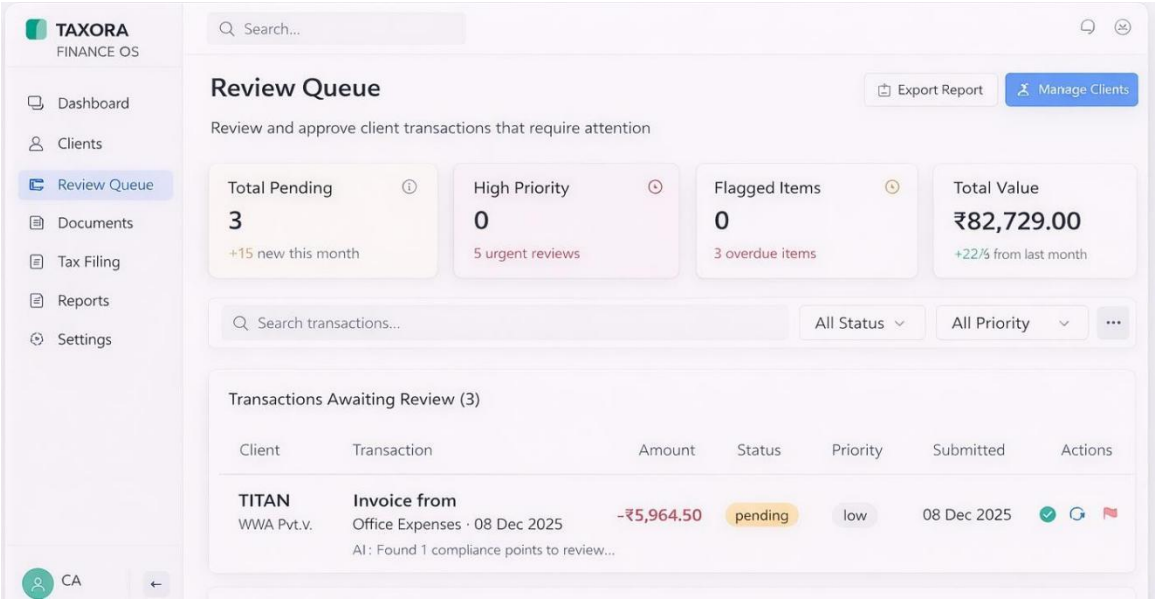

Fig 8.CA Review Queue

This figure presents the invoice review queue where extracted invoices are listed for professional validation. CAs can approve, edit, or reject entries to ensure data accuracy before compliance filing.

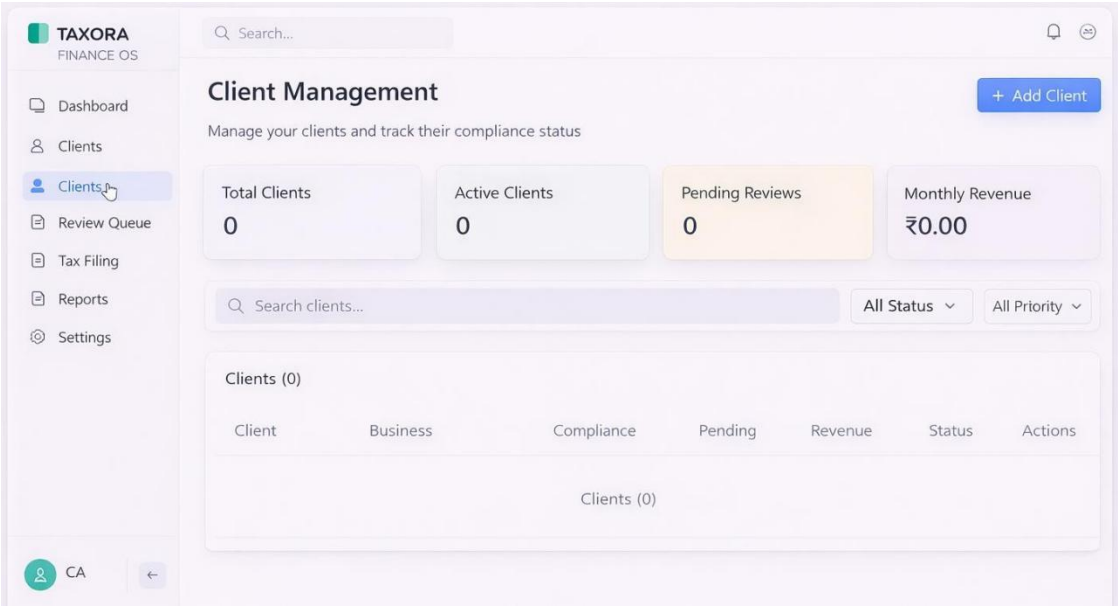

Fig 9.CA Client Management

The client management module displays all linked SME clients and their associated records. It enables CAs to navigate client-specific data, manage requests, and track ongoing accounting activities.

V. Conclusion

This paper presents Taxora, an AI-based accounting and compliance platform designed to address practical challenges faced by SMEs and Chartered Accountants. By combining layout-aware invoice processing using LayoutLMv3 with automated compliance workflows and collaborative dashboards, the system reduces manual effort and enhances accuracy in financial operations. The platform demonstrates the potential of AI-driven solutions to modernize SME accounting and strengthen professional collaboration.

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